



Real-Time Rumor Verification System using AI-Powered Video Surveillance Based Threat Detection



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ABSTRACT

Our country recently faced a mass uprising, during which the regime effectively used rumor propagation to suppress dissent by inciting violence, ambushes, and unrest among citizens. This rapid spread of misinformation highlighted the urgent need for a system capable of detecting rumors in real time. In response, we developed a solution that **combines social media text analysis with real-time surveillance footage** from government CCTV systems. This integrated approach allows us to **detect, categorize, and validate potential threats swiftly**. Even in today's turbulent political landscape, rumors continue to fuel chaos, making our system crucial for mitigating the spread of misinformation.

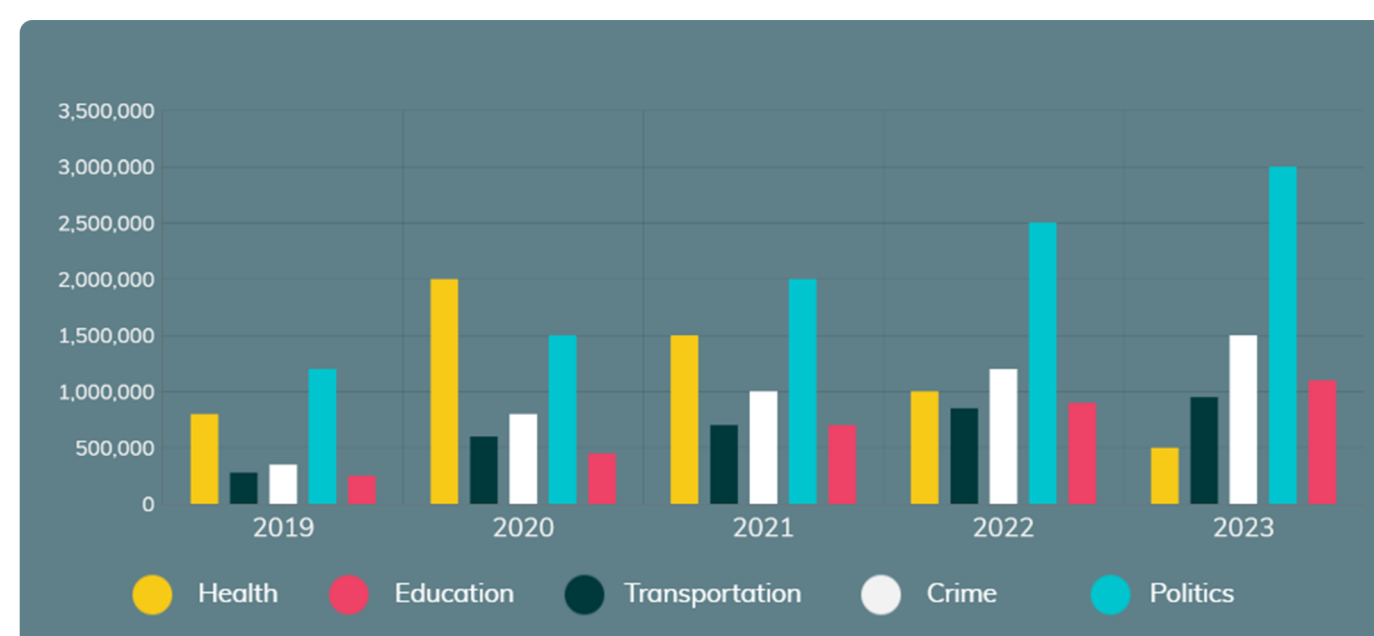


Figure 1: Prevalence of Rumor Types in Bangladesh Over the Last Five Years (2019-2023)

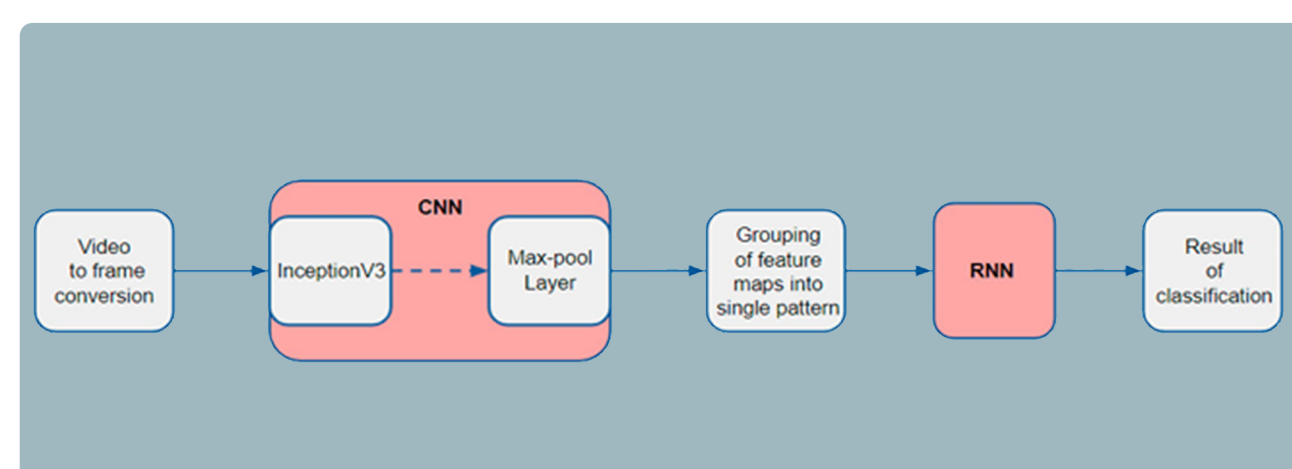


Figure 2: Pipeline for Video Frame Classification Using CNN and RNN Architecture [3]

OBJECTIVES

- Monitor social media posts for potential threats and **categorize them into 16 events**.
- Analyze **real-time CCTV footage** using **ML models** to detect location based activities.
- **Cross-reference results** from social media analysis and CCTV footage through a data fusion approach, aligning event types, time, and location.
- Assign **confidence scores** to validate the authenticity of reported threats by comparing both text and video data.
- Implement **user behavior profiling** to track accounts that frequently post rumors for more rigorous monitoring.
- Enhance public safety by providing **real-time incident validation**.

METHODOLOGY

- Our system **continuously monitors and processes** a fixed number of social media posts per hour, prioritizing them based on **two main factors**: (i) the **engagement rate** (likes, comments, shares) and (ii) the **confidence scores** of the accounts posting.
- We classify events into **16 predefined categories**: Abuse, Burglary, Explosion, Shooting, Fighting, Shoplifting, Road Accidents, Arson, Robbery, Stealing, Assault, Vandalism[3], added with **Police Brutality, Riot/Protests, Road Blockade, and Weapon Display**, plus one "safe" category indicating none of these events.
- If a post identifies a threat (e.g., "Robbery at X"), we verify it against CCTV footage to check for **matching anomalies**.
- Accounts start with a **100% confidence score**, which decreases for sharing rumors.
- Our **fusion module** links social media posts to CCTV footage **based on location and time**. After video classification, the results are sent back to the fusion module for further analysis.

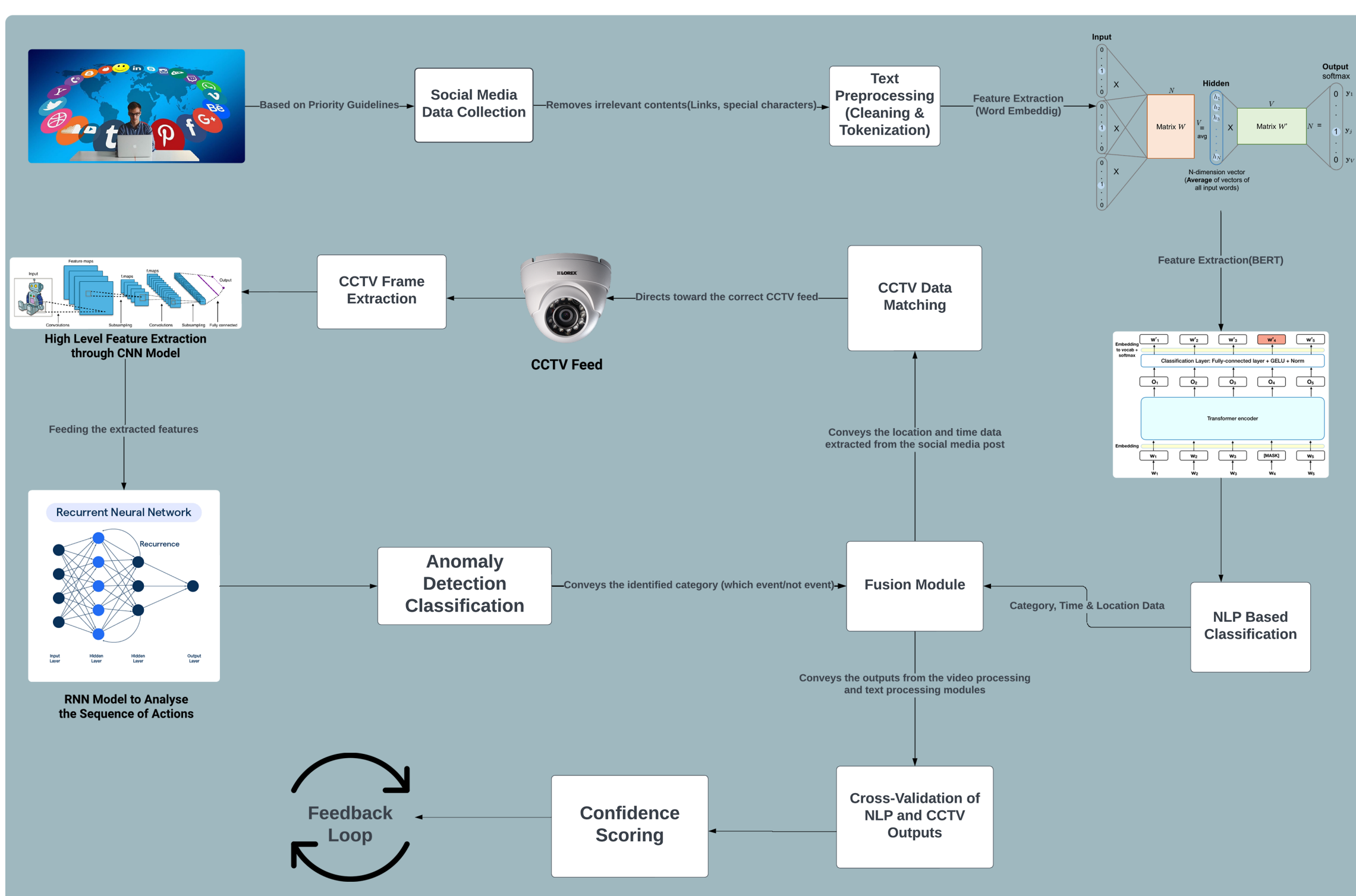


Figure 3: Integrated Framework for Social Media and CCTV-Based Anomaly Detection

FEASIBILITY

- **Technical Viability**: Utilizes pre-trained models such as **VGG-16** for efficient, real-time security analysis.
- **Cost-Effectiveness**: Leveraging existing models reduces development costs and focuses on essential data, minimizing storage requirements.
- **Scalability**: Compatible with various hardware configurations and easily integrates with current CCTV infrastructures.
- **Infrastructure Utilization**: Employs existing governmental CCTV setups, eliminating additional infrastructural expenses.
- **Reduced Human Resource Dependency**: The system's automation reduces the necessity for continuous manual monitoring.

EXPECTED OUTCOMES

- **Accurate Incident Verification**: Enhancing the precision of threat detection by correlating social media analysis with CCTV footage.
- **Misinformation Reduction**: Minimizing the spread of rumors by distinguishing real threats from false reports.
- **Efficient Resource Use**: Optimizing the allocation of security resources through automated verification processes.
- **Adaptive Learning**: Continuously improving the accuracy of the system's predictive models using validated data.
- **Public Safety Improvement**: Contributing to public safety by providing timely and accurate incident validation.

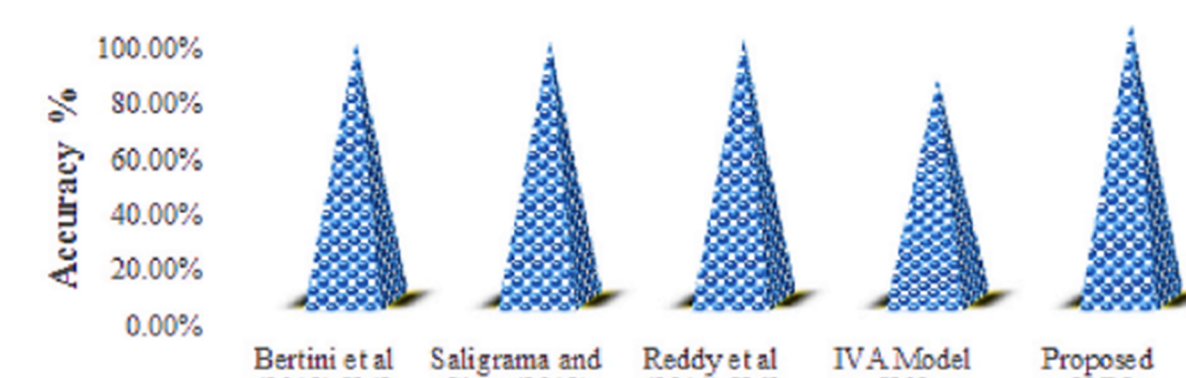


Figure 4: Comparison of Classification Accuracy Across Different Models [2]

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- [3] Virender Singh, Swati Singh, and Pooja Gupta. **Real-time anomaly recognition through cctv using neural networks**. Procedia Computer Science, 173:254–263, 2020. International Conference on Smart Sustainable Intelligent Computing and Applications under ICITETM2020.